

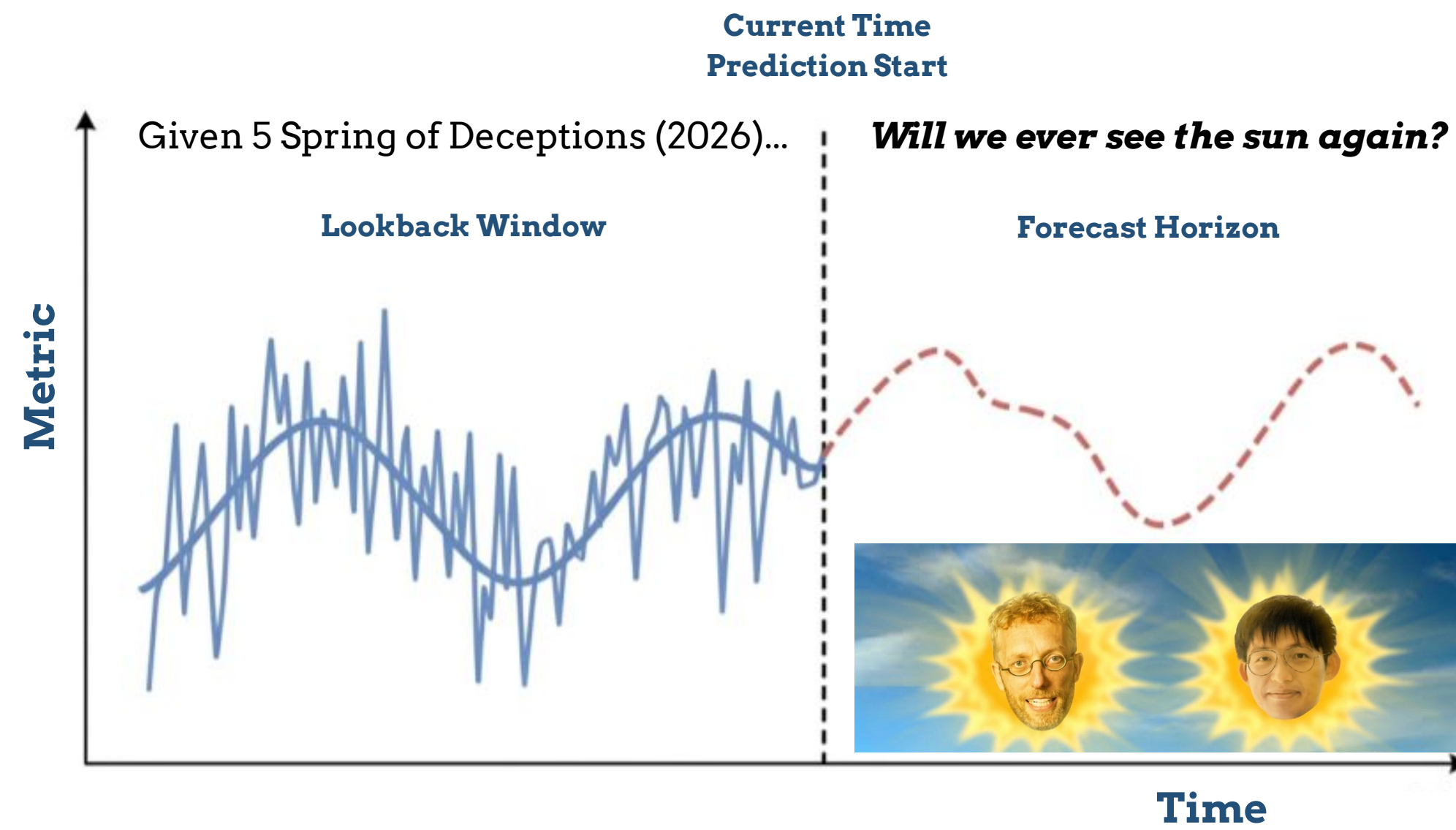
# Exploring PatchTST: A Transformer Approach to Time Series Forecasting

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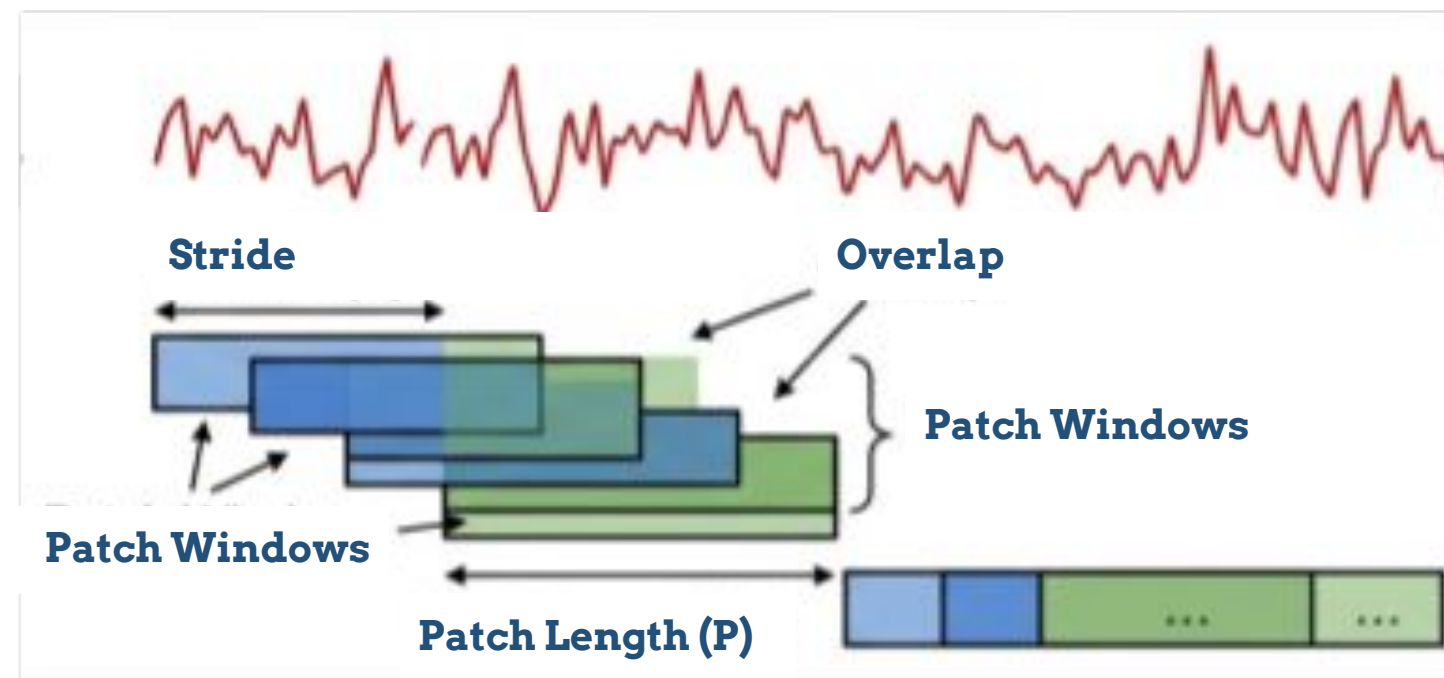
## Introduction



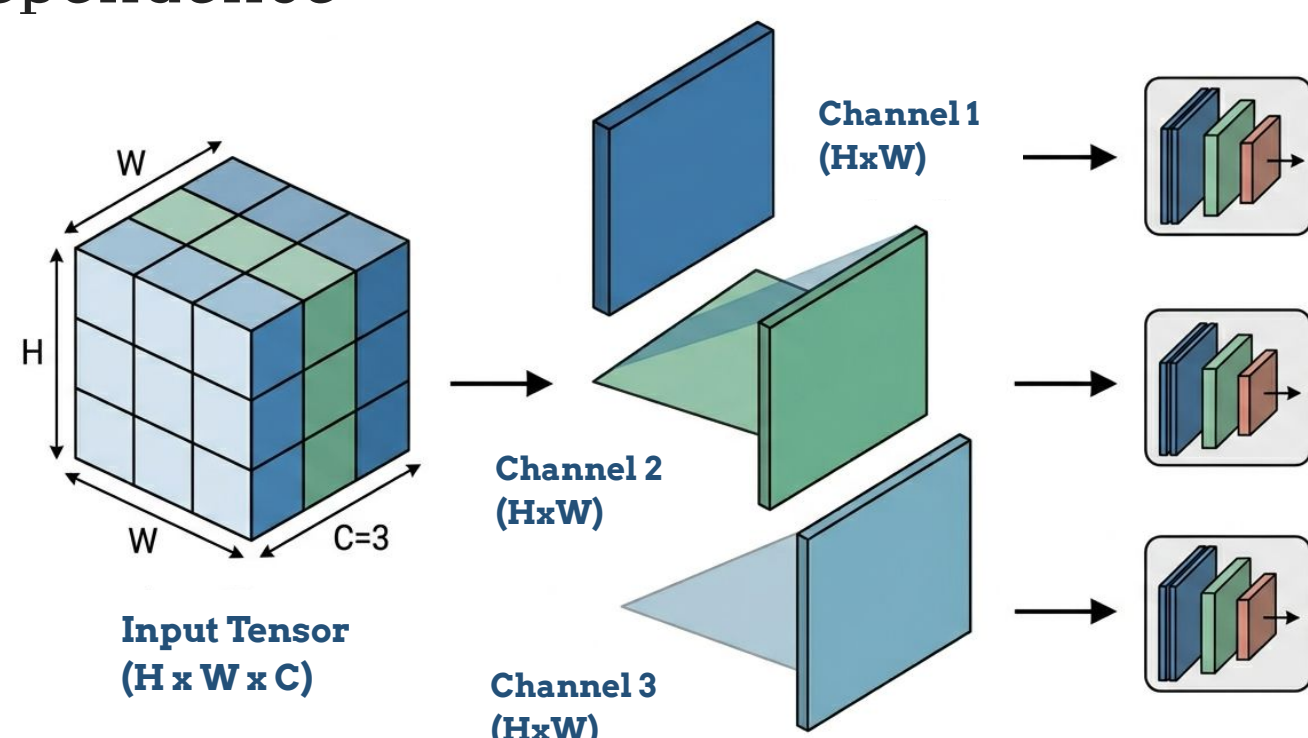
## Background and Motivation

- Transformer models applied to time series tasks were being beat by linear models [2] They are different from traditional NLP tasks
- Our goal is to show that with patching and channel independence, we can show that transformer models outperform linear models on time series tasks

### 1. Patching

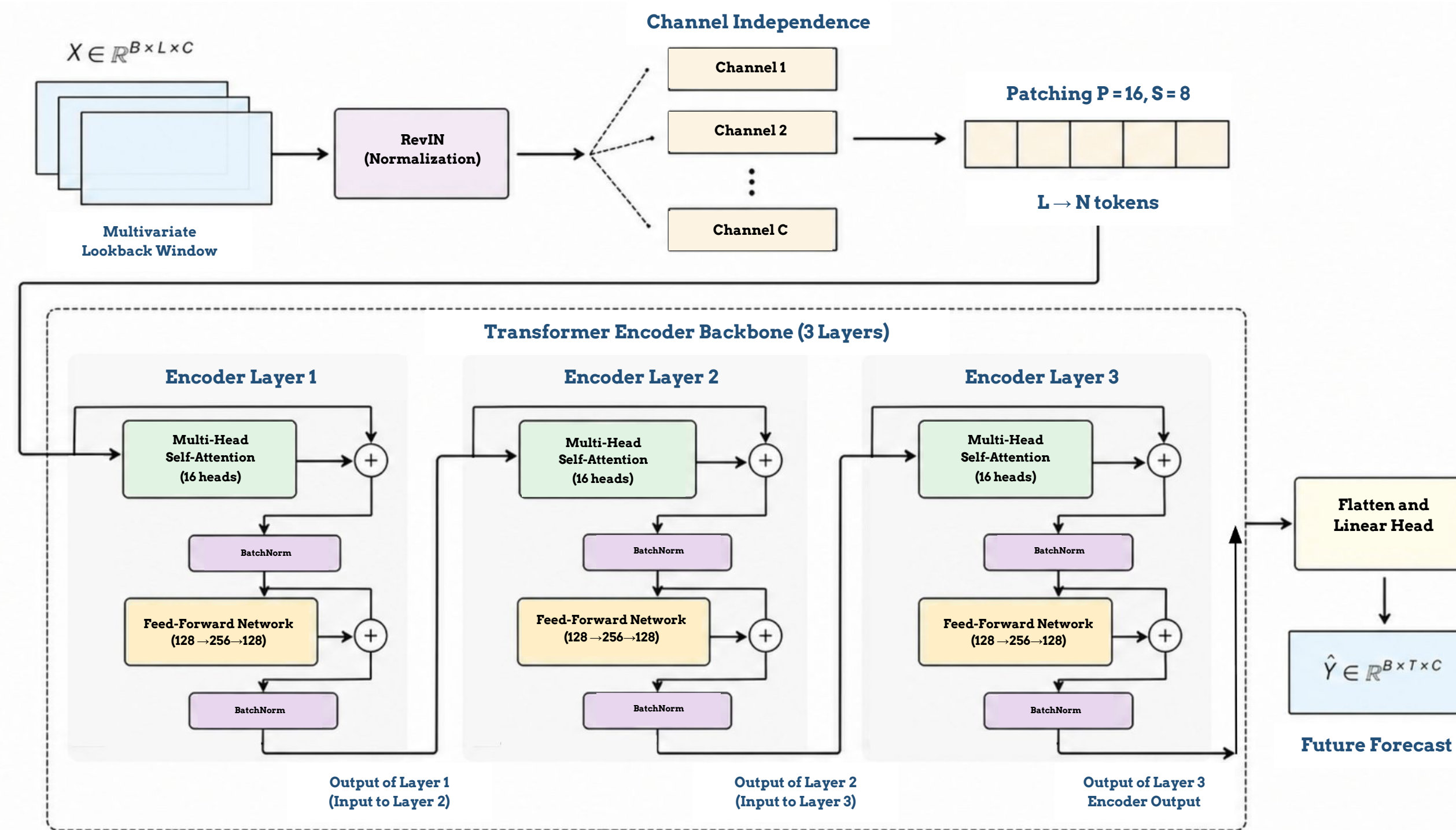


### 2. Channel Independence

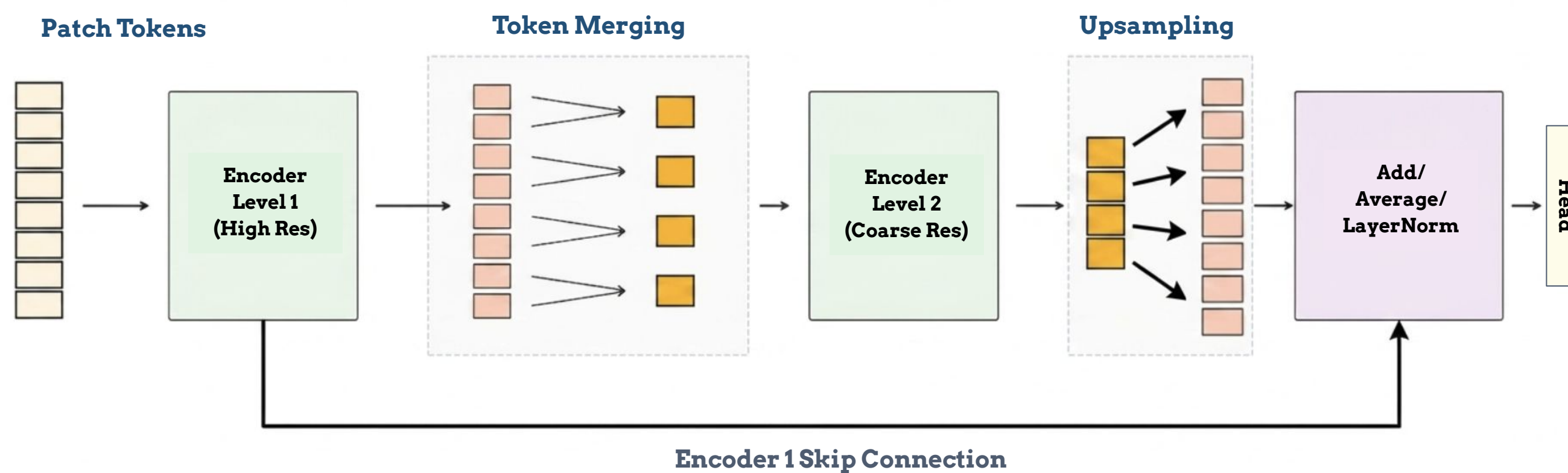


## Methods

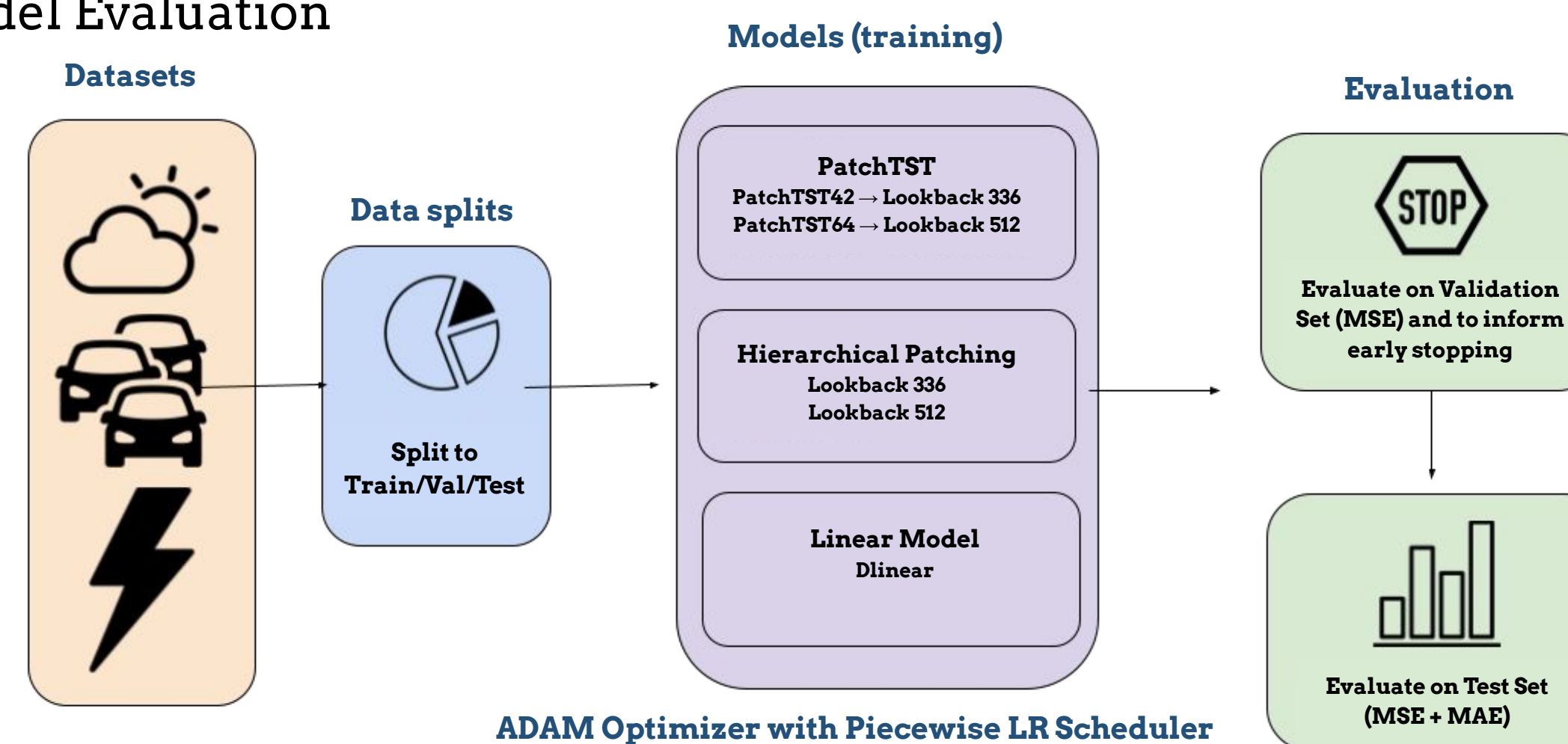
### Architecture



### Hierarchical Patching Independent Exploration (HPatchTST)

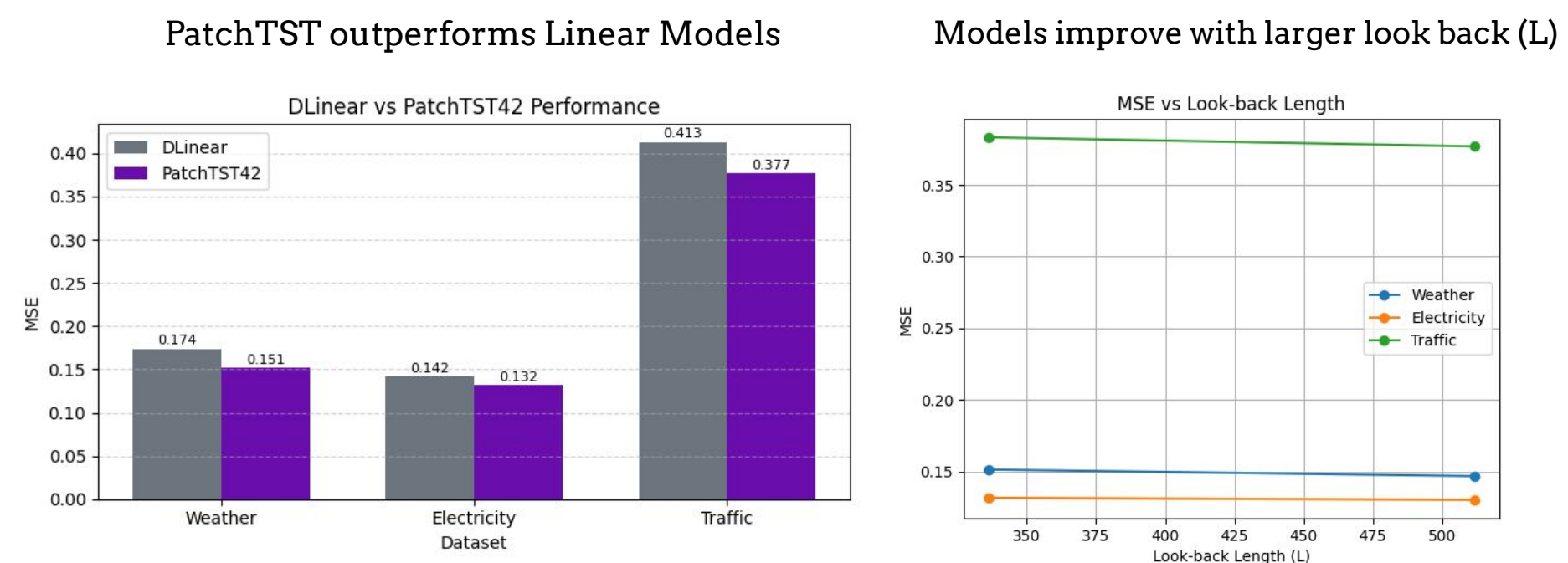


### Model Evaluation



## Results

|       | Model               | Dataset       |               |               |               |               |               |
|-------|---------------------|---------------|---------------|---------------|---------------|---------------|---------------|
|       |                     | Weather       |               | Electricity   |               | Traffic       |               |
|       | Pred Len = 96       | MSE           | MAE           | MSE           | MAE           | MSE           | MAE           |
| Ours  | dLinear             | 0.1740        | 0.2349        | 0.1416        | 0.2398        | 0.4129        | 0.2888        |
|       | PatchTST/42 (L=336) | 0.1513        | 0.1997        | 0.1317        | 0.2258        | 0.3830        | 0.2618        |
|       | PatchTST/64 (L=512) | <b>0.1468</b> | <b>0.1963</b> | 0.1302        | 0.2249        | <b>0.3767</b> | 0.2602        |
|       | HPatch/42 (L=336)   | 0.1495        | 0.1970        | 0.1319        | 0.2254        | 0.3822        | 0.2643        |
|       | HPatch/64 (L=512)   | 0.1478        | 0.1988        | 0.1301        | 0.2238        | <b>0.3767</b> | 0.2627        |
| Paper | PatchTST/42 (L=336) | 0.1520        | 0.1990        | 0.1300        | <b>0.2220</b> | 0.3670        | 0.2510        |
|       | PatchTST/64 (L=512) | 0.1490        | 0.1980        | <b>0.1290</b> | <b>0.2220</b> | 0.3670        | <b>0.2490</b> |

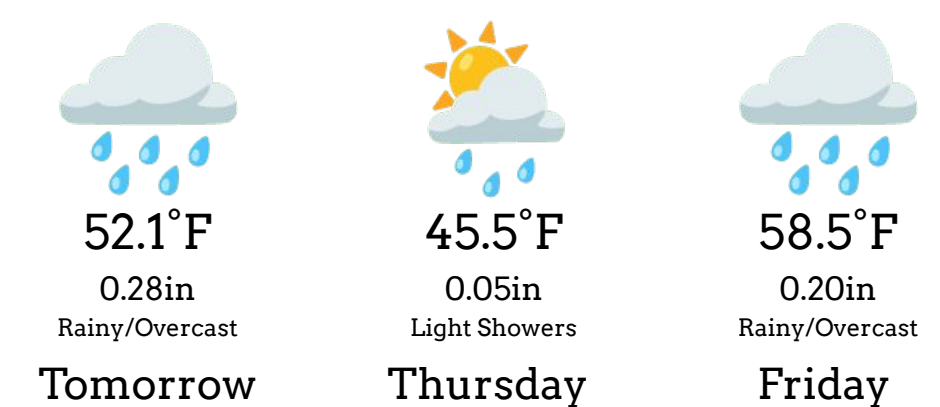


## Conclusion

### Takeaways:

- PatchTST (and HPatchTST) perform better than linear models
- Reduction in time and space complexity ( $N = L/S$ )
- Hierarchical model gains are insignificant from base PatchTST
  - Marginal gains in specific datasets, warrants further study

### So will we see the sun?



No, we won't :(

### Future work:

- Modelling cross channel dependencies
- Adaptive patching

## References:

- [1] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2022. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *arXiv:2211.14730*.
- [2] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *arXiv preprint arXiv:2205.13504*, 2022.